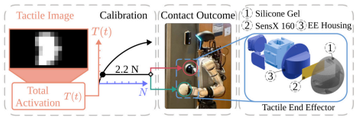
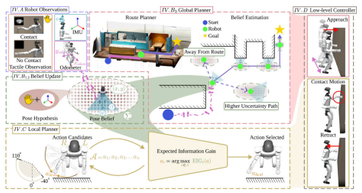
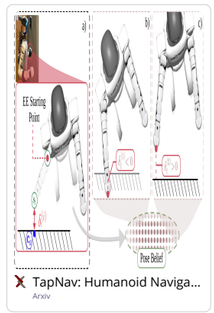
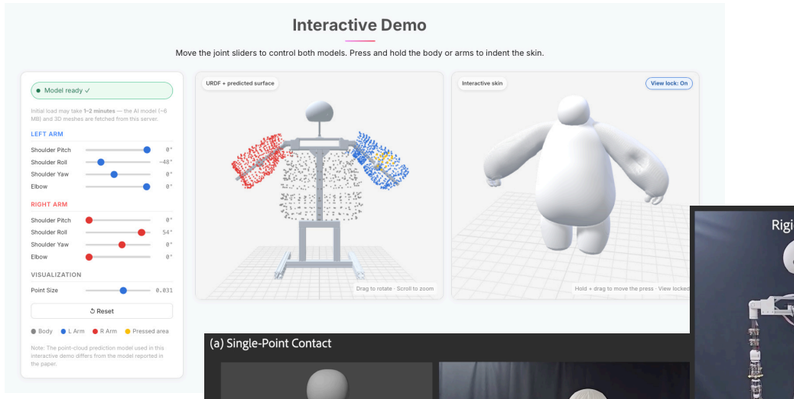
Research Mood Board









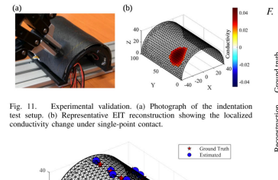


Fig. 11. Experimental validation. (a) Photograph of the indentation test setup. (b) Representative EIT reconstruction showing the localized conductivity change under single-point contact.

F. Proof-of-concept on a complex shape

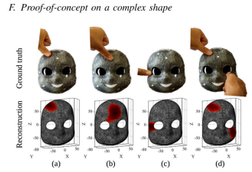
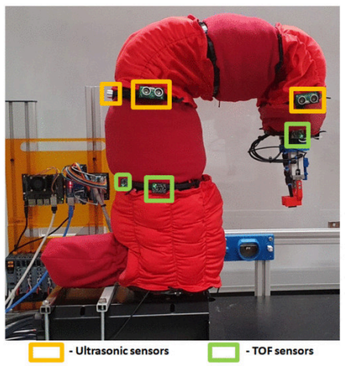


Fig. 13. Example interactions with the iCub face sensor prototype. In each column, the ground truth (photography) is presented above the resulting EIT reconstruction. See the supplementary video for more information.

Finally, we explored the feasibility of applying the proposed conformal EIT sensing concept to a more complex freeform geometry. For this purpose, the front surface of the iCub robot's face was chosen. Unlike the planar and cylindrical prototypes used for quantitative evaluation, the iCub-face-





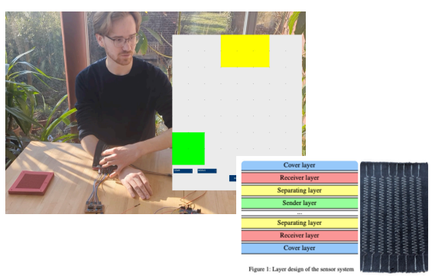
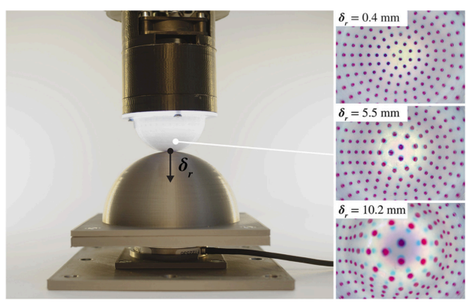


Figure 1: Layer design of the sensor system



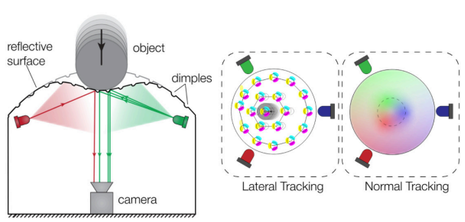


Fig. 2. ShadowTac working principle. Light is emitted around the reflective membrane. The dimple features embedded in the membrane create distinctive shadows from which the lateral displacement can be tracked. Retrographic methods can reconstruct the normal displacement field.

